GCSE Physics
Exam Walkthrough Pack

Step-by-step worked exam-style questions with full explanations

Original JDScience exam-style questions — not copied from past papers

Introduction

This Physics Exam Walkthrough Pack is designed to help you understand how to approach GCSE and IGCSE exam-style questions. Every question in this booklet is an original JDScience item written to match common specification skills — it is not copied from AQA, Edexcel, OCR or other past papers.

How to use this pack: read each question, cover the walkthrough, and try answering in exam conditions first. Then read what the question is asking, follow the step-by-step thinking, compare your answer with the model answer, and study the mark breakdown to see exactly how marks are awarded.

Command words tell you what type of answer is required. Section 1 explains the most common GCSE science command words. Always underline the command word before you start writing.

Showing working matters: in calculations, examiners award method marks even when the final answer is wrong. In longer answers, a clear logical sequence helps you secure every available mark.

Examiners award marks for correct science linked to the question. Keywords from the specification matter, but a correct idea in your own words can still score. Extended questions often use level-based marking: plan before you write, use paragraphs, and link ideas.

To move from Grade 4/5 to Grade 7/8/9: use precise terminology, explain mechanisms (not just names), include data or examples where asked, and check units and significant figures in calculations. For six-mark questions, a short plan and linked paragraphs beat a long unordered list.

Section 1: Command word guide

State

What you must do: Give a short, factual answer — often one word, number or phrase. No explanation is required unless the question also asks you to explain.

Common mistake: Writing a long paragraph when a single fact is enough, or giving an explanation when only a fact was asked for.

Model sentence starter: The [quantity/name] is …

Give

What you must do: Provide the information requested. Similar to state, but may need a short phrase or named example.

Common mistake: Listing unrelated facts that do not answer the question.

Model sentence starter: One example is … / The value is …

Describe

What you must do: Say what happens or what something is like. Focus on observations, patterns or features. You do not need to say why.

Common mistake: Explaining causes when the command word is only describe.

Model sentence starter: First … then … / The pattern shows …

Explain

What you must do: Give reasons why something happens. Link cause and effect using science ideas and key terms.

Common mistake: Describing what happens without saying why, or using everyday language instead of science.

Model sentence starter: This happens because … which means …

Calculate

What you must do: Work out a numerical answer. Show your working, include units, and give the final value to the required precision.

Common mistake: Missing units, no working, or using the wrong equation.

Model sentence starter: Use … = … / Substitute: …

Compare

What you must do: Identify similarities and/or differences between two or more things. Refer to both sides.

Common mistake: Only describing one item, or listing features without comparing.

Model sentence starter: Both … however … / X has … whereas Y …

Evaluate

What you must do: Weigh up evidence or options, consider strengths and weaknesses, and reach a supported judgement.

Common mistake: Only listing advantages with no conclusion or balanced comment.

Model sentence starter: An advantage is … A limitation is … Overall …

Suggest

What you must do: Apply your knowledge to a new situation. Your answer should be sensible and scientifically plausible.

Common mistake: Repeating textbook facts that do not fit the context given.

Model sentence starter: A possible reason is … / This could be because …

Justify

What you must do: Give evidence or reasons that support a choice, conclusion or statement already made.

Common mistake: Restating the claim without giving supporting science.

Model sentence starter: This is supported by … because …

Determine

What you must do: Find out a value or conclusion using data, a graph or a calculation. Show how you reached your answer.

Common mistake: Reading a graph incorrectly or not showing how the value was obtained.

Model sentence starter: From the graph … / Using the data …

Analyse

What you must do: Break information into parts, interpret patterns or trends, and explain what the data shows.

Common mistake: Describing data without interpreting it, or ignoring anomalies.

Model sentence starter: The data shows that … / As X increases, Y …

Section 2: Energy

QUESTION 1

State the unit of energy.

QUESTION TYPE

Topic: Energy | **Command word:** state | **Marks:** 1 | **Skill tested:** Knowledge recall

WHAT THE QUESTION IS ASKING

The question tests whether you know the SI unit used for energy.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

The unit of energy is the joule (J).

MARK BREAKDOWN

Mark 1: Correct unit named as joule or J.

COMMON MISTAKES

- Writing watts (W) instead of joules.
- Spelling 'joules' without the symbol when asked for the unit.

EXAMINER TIP

Energy is measured in joules; power is measured in watts.

QUESTION 2

Give one example of a renewable energy resource.

QUESTION TYPE

Topic: Energy | **Command word:** give | **Marks:** 1 | **Skill tested:** Knowledge recall

WHAT THE QUESTION IS ASKING

You must name one energy resource that is naturally replenished.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

One example is solar energy.

MARK BREAKDOWN

Mark 1: Any valid renewable named, e.g. solar, wind, hydroelectric, tidal, geothermal, or biomass.

COMMON MISTAKES

- Naming a fossil fuel such as coal.
- Naming nuclear fuel without noting it is non-renewable.

EXAMINER TIP

Learn a short list of renewables and one non-renewable for quick comparison questions.

QUESTION 3

State what is meant by a closed system in energy transfers.

QUESTION TYPE

Topic: Energy | **Command word:** state | **Marks:** 1 | **Skill tested:** Knowledge recall

WHAT THE QUESTION IS ASKING

Define the idea of a closed system in the context of energy.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

A closed system is one where no energy is transferred to or from the surroundings.

MARK BREAKDOWN

Mark 1: States that no energy enters or leaves the system.

COMMON MISTAKES

• Saying no matter can enter or leave (that describes a closed system in chemistry, not the energy definition needed here).

• Confusing closed with isolated systems at A-level.

EXAMINER TIP

In GCSE physics, closed system usually means the total energy stays the same.

QUESTION 4

Describe the energy transfer when a student lifts a book from the floor to a shelf.

QUESTION TYPE

Topic: Energy | **Command word:** describe | **Marks:** 2 | **Skill tested:** Energy transfers

WHAT THE QUESTION IS ASKING

Describe what happens to energy during this transfer without explaining why in detail.

STEP-BY-STEP THINKING

1. Identify the initial and final energy stores.
2. Name the method of transfer (work done by a force).
3. State energy gained by one store and supplied from another.
4. Use correct energy store names.
5. Check both stores are mentioned.

MODEL ANSWER

Chemical energy in the student's muscles decreases and gravitational potential energy of the book increases. Energy is transferred mechanically by the upward force doing work on the book.

MARK BREAKDOWN

Mark 1: Chemical store of muscles identified as decreasing.

Mark 2: Gravitational potential energy of book identified as increasing.

COMMON MISTAKES

• Saying energy is created.

• Using 'kinetic' only without mentioning the work done against gravity.

EXAMINER TIP

Always name both the store that decreases and the store that increases.

QUESTION 5

Explain why a car's brakes get hot when the car slows down.

QUESTION TYPE

Topic: Energy | **Command word:** explain | **Marks:** 2 | **Skill tested:** Energy dissipation

WHAT THE QUESTION IS ASKING

Give the reason linked to energy transfer when braking.

STEP-BY-STEP THINKING

1. Identify the kinetic energy store involved.
2. State where energy is transferred to.
3. Link heating to friction.
4. Use the idea of wasted or dissipated energy.
5. Check the explanation answers why, not just what happens.

MODEL ANSWER

The brakes get hot because kinetic energy of the car is transferred to the thermal store of the brakes and surroundings through friction, which dissipates useful energy as heat.

MARK BREAKDOWN

Mark 1: Kinetic energy decreases or is transferred.

Mark 2: Thermal energy increases in brakes due to friction / dissipation.

COMMON MISTAKES

- Saying energy is destroyed.
- Only stating brakes touch without mentioning energy stores.

EXAMINER TIP

Use 'transferred to the thermal store' rather than 'lost'.

QUESTION 6

Compare the energy changes in a pendulum at the highest point and at the bottom of its swing.

QUESTION TYPE

Topic: Energy | **Command word:** compare | **Marks:** 2 | **Skill tested:** Energy stores

WHAT THE QUESTION IS ASKING

Identify similarities and differences in energy stores at two positions.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.
5. Check units, keywords and that your final sentence answers the command word.

MODEL ANSWER

At the highest point the pendulum has maximum gravitational potential energy and minimum kinetic energy. At the bottom, gravitational potential energy is minimum and kinetic energy is maximum. In both positions, if air resistance is ignored, total energy is conserved.

MARK BREAKDOWN

Mark 1: Correct stores compared at both positions (GPE high/low, KE low/high).

Mark 2: Similarity or conservation comment, or clear contrast of both positions.

COMMON MISTAKES

- Only describing one position.
- Ignoring kinetic energy at the bottom.

EXAMINER TIP

Compare command words need features of both situations.

QUESTION 7

Explain why insulating a house reduces heating costs.

QUESTION TYPE

Topic: Energy | **Command word:** explain | **Marks:** 3 | **Skill tested:** Efficiency and insulation

WHAT THE QUESTION IS ASKING

Link insulation to reduced energy transfer and cost.

STEP-BY-STEP THINKING

1. State what insulation reduces (thermal transfer).
2. Explain less energy leaves the home.
3. Link to heater switching on less often.
4. Connect to lower fuel/electrical use.
5. Use correct terminology: thermal store, transfer, efficiency.

MODEL ANSWER

Insulation reduces the rate of energy transfer by heating to the surroundings. Less energy must be supplied by the heating system to maintain the same temperature, so less fuel or electricity is used and heating costs fall.

MARK BREAKDOWN

Mark 1: Insulation reduces energy transfer to surroundings.

Mark 2: Less energy input needed to maintain temperature.

Mark 3: Lower energy use reduces cost.

COMMON MISTAKES

- Saying insulation creates energy.
- Only stating 'keeps heat in' without energy transfer language.

EXAMINER TIP

Link every efficiency answer to reduced wasted energy transfer.

QUESTION 8

A kettle transfers 180 kJ of energy to water in 3 minutes. Describe how you would calculate its power and state what extra information you already have.

QUESTION TYPE

Topic: Energy | **Command word:** describe | **Marks:** 3 | **Skill tested:** Power

WHAT THE QUESTION IS ASKING

Outline the calculation method using given data.

STEP-BY-STEP THINKING

1. Identify the science idea the question is testing.
2. Pick out key data or keywords in the question stem.
3. Decide which equation, rule or process applies.
4. Build the answer logically in the order the marks expect.